Review Pulse v2

Aspect-Based Sentiment Analysis with Attention-Based Deep Learning

DLE602 Deep Learning - Assessment 2: Project Proposal

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The problem - why one label is not enough

"The food was **great** but the service was **slow**."

- One review → **two opposite opinions**, one per aspect
- Sentence-level models collapse it to a single 😐 : *food* **good** / *service* **bad** is lost
- Our A1 N-gram classifier and the Zhao et al. (2018) CNN both do exactly this

A business reading "neutral" learns nothing. "Food good, service bad" is actionable.

What the literature shows

CNN	ATAE-LSTM	BERT-for-ABSA	LLM / neural discovery
2018	2016	2019	2023-2026
deep but flat	attention	contextual	frontier

CNN → ATAE-LSTM → BERT-for-ABSA → LLM / neural discovery

- One limitation - a single label per text - solved step by step
- The frontier is **heavy and LLM-driven**; few pair a *light, explainable* model with faithful attention

We are here explainable · low-compute · aspect-level

Our approach

(review, aspect) → **ATAE-LSTM** + DistilBERT → per-aspect sentiment → **attention heatmap**

- **Two models compared:** aspect-aware attention-LSTM vs fine-tuned DistilBERT
- **Interpretability is the differentiator** - we show *why*, not just the label
- Optional neural **Topic Modelling** to discover aspects (scoped as stretch)

SemEval-2014: Restaurants + Laptops

accuracy · macro-F1 · attention

Plan & risks - feasible and de-risked

Modules 8 → 12

data + baseline → ATAE-LSTM → DistilBERT → interpretability + Streamlit demo → eval + report

Risk	Mitigation
Small data → overfitting	dropout · early stopping · transfer learning
Heavy fine-tuning	DistilBERT on free Colab GPU
Scope creep	extraction & topic modelling are **gated stretch goals**

Critical success factors: pre-annotated data · light compute · working LSTM fallback

Expected contribution

Aspect-level	Explainable	Live demo
sentiment per thing	attention heatmap	Streamlit, Assessment 3

From one score per review to a **transparent, per-aspect read** - built and demoed in Assessment 3.

References (APA)

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